

CHRISTOPHER BITLER

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SENIOR SOFTWARE ENGINEER

*BUILDING RELIABLE APPLICATIONS, SCALABLE SYSTEMS, EFFICIENT ENGINEERING WORKFLOWS,
AND PRACTICAL AI SOLUTIONS FOR COMPLEX BUSINESS PROBLEMS*

PROFESSIONAL EXPERIENCE

Opencash (Acquired by InvestView) | Haverford, PA - Remote (September 2025–Present)

SOFTWARE ENGINEER

Platform Architecture, Security & Reliability

- Separated PII tokenization, market data, and job management from the main backend into dedicated NestJS/TypeScript microservices, improving separation of concerns and limiting unnecessary access to sensitive data.
- Researched whether to build SSN tokenization internally or use an outside provider, comparing infrastructure, engineering time, required functionality, and cost before recommending EnigmaVault at approximately \$50 per month.
 - Built the service using TDD and deterministic tokenization so duplicate SSNs could still be detected, restricted de-tokenization to authorized staff, and recorded all activity in WORM-protected S3 audit logs; had the service running in staging within one week.
- Designed the weekly scratch-off system so Opencash could prove after the fact that results had not been changed and could reproduce the same results if questioned by FINRA, the SEC, or another auditor.
 - Used hash-chained files stored in WORM-protected S3 and combined entropy from AWS KMS, NIST, and DRand so changes to prior data would stop processing and the source of the random seed could be independently verified.
- Reworked scheduled-job processing after finding that EventBridge could schedule jobs but could not adequately retry failures or stop later jobs when an earlier step failed.
 - Built an AWS Step Functions/ECS workflow with standardized success/failure reporting and exponential-backoff retries.
 - Used the same framework for the roughly eight-job weekly scratch-off process, allowing processing to stop on a critical failure, alert the on-call engineer, and resume from the failed step after the problem was corrected; significantly reduced incomplete jobs.

Integrations, Delivery & AI

- Integrated the platform with clearing firm RQD to support account reconciliation, trade orders, and customer reward transfers.
 - Used webhooks and daily SFTP reports to reconcile Opencash records with clearing-firm data and added error handling around trade-order processing and updates.
- Removed a manual TypeORM database-migration step that could be missed, run against the wrong environment, or leave the application and database out of sync.
 - Used AWS CodeBuild as a GitHub Actions runner inside the VPC to run migrations automatically during staging and production deployments.
- Built an OpenAI API-based support chatbot so users could get answers about the application and general trading terminology without contacting support.
 - Compared available AI options and used OpenAI Structured Outputs, Guardrails, and prompt instructions to keep the chatbot from providing financial advice; reduced support calls and emails.
- Simplified a poorly documented React Native/Expo release process that was prone to mistakes by creating a GitHub Action that generated a pull request showing all changes since the prior release.
 - Once approved and merged, the workflow automatically started the Expo release, reducing release errors and making the process much easier for newer engineers to learn and run.

Intuit | Mountain View, CA (May 2020–April 2025)

SOFTWARE ENGINEER 2 (MAY 2021–APRIL 2025) / SOFTWARE ENGINEER (MAY 2020–MAY 2021)

Earned 22 peer-nominated Spotlight Awards over four years for engineering contributions across lending, payments, and cryptocurrency products within QuickBooks Capital and QuickBooks Online.

Product, AI & Full-Stack Development

- Transformed QuickBooks Capital’s underwriting process by implementing a TypeScript/React generative AI solution that analyzed loan applications, identified issues, and provided remediation steps, reducing review time and allowing underwriters to process more applications without sacrificing review quality.
- Developed an AI-powered Node.js/TypeScript migration tool that automatically converted JavaScript to TypeScript, eliminating manual conversion work and saving engineering time; one of four teams recognized with an organization-level award for engineering efficiency.
- Led the UI implementation for QuickBooks Crypto’s buy/sell/hold feature, using TypeScript, React, React Query, and Kotlin to give customers the ability to purchase, hold, or sell cryptocurrency through custodial wallet partners using QuickBooks Checking funds.
- Redesigned portions of the QuickBooks Line of Credit acceptance flow in TypeScript/React, improving input error handling and reducing support tickets while supporting efforts to increase loan approvals and volume.

Architecture, APIs & Developer Productivity

- Reduced production issues from breaking API changes by developing a Groovy/Jenkins build pipeline that automatically generated TypeScript definitions from OpenAPI schemas; earned a Spotlight Award for the solution and for presenting it to the development team.
- Connected QuickBooks UI flows with multiple payment and lending microservices through Apollo GraphQL, allowing user, payment, and loan data to be queried and updated across services.
- Increased line-of-credit signups among customers who declined term-loan offers by implementing Java/Spring Boot backend logic that triggered targeted marketing emails through Camunda.

CO-OP EXPERIENCE

Intuit | Mountain View, CA – Software Engineer Intern (May 2019–December 2019)

Datto | Rochester, NY – Software Engineering Intern (January 2017–August 2017 and May 2018–August 2018)

TECHNICAL SKILLS / AREAS OF EXPERTISE

Languages: TypeScript | JavaScript | Java | Kotlin | HTML/CSS

Frontend: React | React Native | React Query | Redux | Next.js | Tailwind CSS

Backend & APIs: Node.js | NestJS | Spring Boot | GraphQL/Apollo | REST/OpenAPI | Microservices | TypeORM

Cloud, Containers & CI/CD: AWS (ECS, S3, Step Functions, CodeBuild, Cognito, IAM, CloudWatch) | GitHub Actions | Jenkins | Terraform | Docker | Kubernetes

Data, Testing & Observability: PostgreSQL | MySQL | MongoDB | TDD | Jest | Playwright | React Testing Library | Grafana/Prometheus | Splunk

Engineering Practices & Architecture: System Architecture | API Design | Security Engineering | Performance Optimization | Developer Tooling & Automation

AI Engineering: Generative AI | OpenAI API | Structured Outputs | Guardrails | OpenAI Codex

EDUCATION / CERTIFICATIONS

B.S. Computer Science, Rochester Institute of Technology

Developing Generative AI Applications with Python | IBM | Certificate ID: [09a32664e748450c9c6e28dff07071db](#)

Verified Certificate for Introduction to Generative AI | IBM | Certificate ID: [c47f5313853c4ca9bd0463e05b38258e](#)